

Section B				
9	(a)	(i)	1. Diffraction refers to the <u>bending or spreading out of waves when they travel through a small opening or when they pass round a small obstacle.</u>	B1
			2. Interference refers to the <u>superposing of two or more coherent waves to produce regions of maxima and minima</u> in space, according to the principle of superposition	B1
			3. Coherence refers to having a <u>constant phase difference</u> (and same frequency) (between waves/sources/particles).	B1
		(ii)	Any two of the following: 1. The waves must overlap to produce regions of maxima and minima. 2. The sources must be coherent . 3. The waves must have the same amplitude or approximately the same amplitude . 4. The waves must be unpolarised or with the same plane of polarisation (for transverse waves).	B1 B1
	(b)	(i)	Since the two radio waves <u>emitters are in phase</u> , along centre-line and <u>path difference is always zero</u> , hence <u>constructive interference</u> always occurs.	B1
		(ii)	Radio waves <u>have long wavelengths</u> , hence the <u>anti-nodal lines will be far apart enough for the ship to differentiate</u>	A1 M1
		(iii)	Since the <u>intensity of each individual wave is inversely proportional to the square of the distance</u> , the <u>intensity of each individual wave will increase</u> as the ship goes nearer, hence the resultant intensity will increase. OR Since the <u>amplitude of each individual wave is inversely proportional to distance</u> , the <u>amplitude of each individual wave will increase</u> as the ship goes nearer, hence the resultant amplitude of	M1

		the superposed wave will increase. As <u>intensity is proportional to the square of the amplitude</u> , intensity increases. Hence, <u>the intensity of the resultant increases as the ship approached the gate.</u>	A1
(c)		$\lambda_1 = \frac{c}{f_1} = \frac{3.0 \times 10^8}{23.5 \times 10^6} = 12.77 \text{ m}$ $\text{Path difference} = \sqrt{(895)^2 + \left(250 + \frac{95}{2}\right)^2} - \sqrt{(895)^2 + \left(250 - \frac{95}{2}\right)^2}$ $= 25.527 \text{ m}$ $\approx 2\lambda_1$ Since the path difference is approximately $2\lambda_1$, the ship is on an anti-nodal line. Note: using $x = \frac{\lambda D}{a}$ will earn no credit.	C1 M1 A1
(d)	(i)	If ship is <u>on the central anti-nodal line</u> , it should detect <u>maximum signals from both frequencies / the maximum signal will be stronger</u> . OR If ship is <u>on wrong anti-nodal line</u> , only <u>one of the frequencies will show a strong signal</u> .	B1
	(ii)	<u>Higher orders of maxima from both frequencies may still coincide/overlap.</u> Hence the ship could still detect maximum signals from both frequencies even though it is not on the central anti-nodal line.	B1
(e)	(i)	Simple harmonic motion	B1
	(ii)	$5.0 \text{ km h}^{-1} = 1.39 \text{ m s}^{-1}$ $v = r\omega$ $v = \frac{2\pi r}{T}$ $T = \frac{2\pi r}{v}$ $= \frac{2\pi(0.225)}{1.39}$ $= 1.02 \text{ s}$	M1 A1
	(iii)	$a_0 = \omega^2 x_0$ $= \left(\frac{2\pi}{1.02}\right)^2 (0.225)$ $= 8.57 \text{ m s}^{-2}$ (Alternatively, use $a = \frac{v^2}{r}$ to solve.)	M1 A1

10	(a)	(i)	S shown on the peak	B1
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